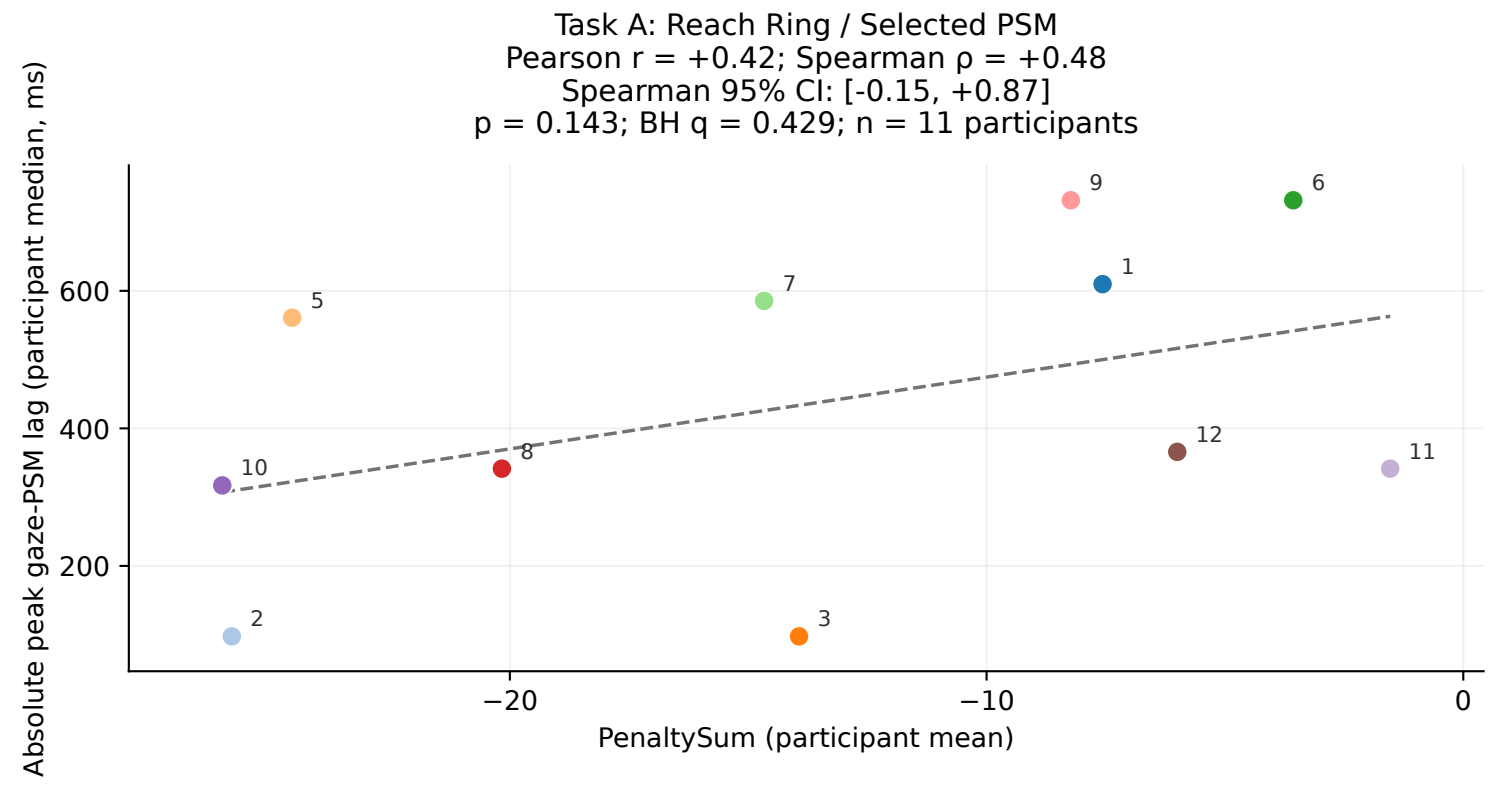
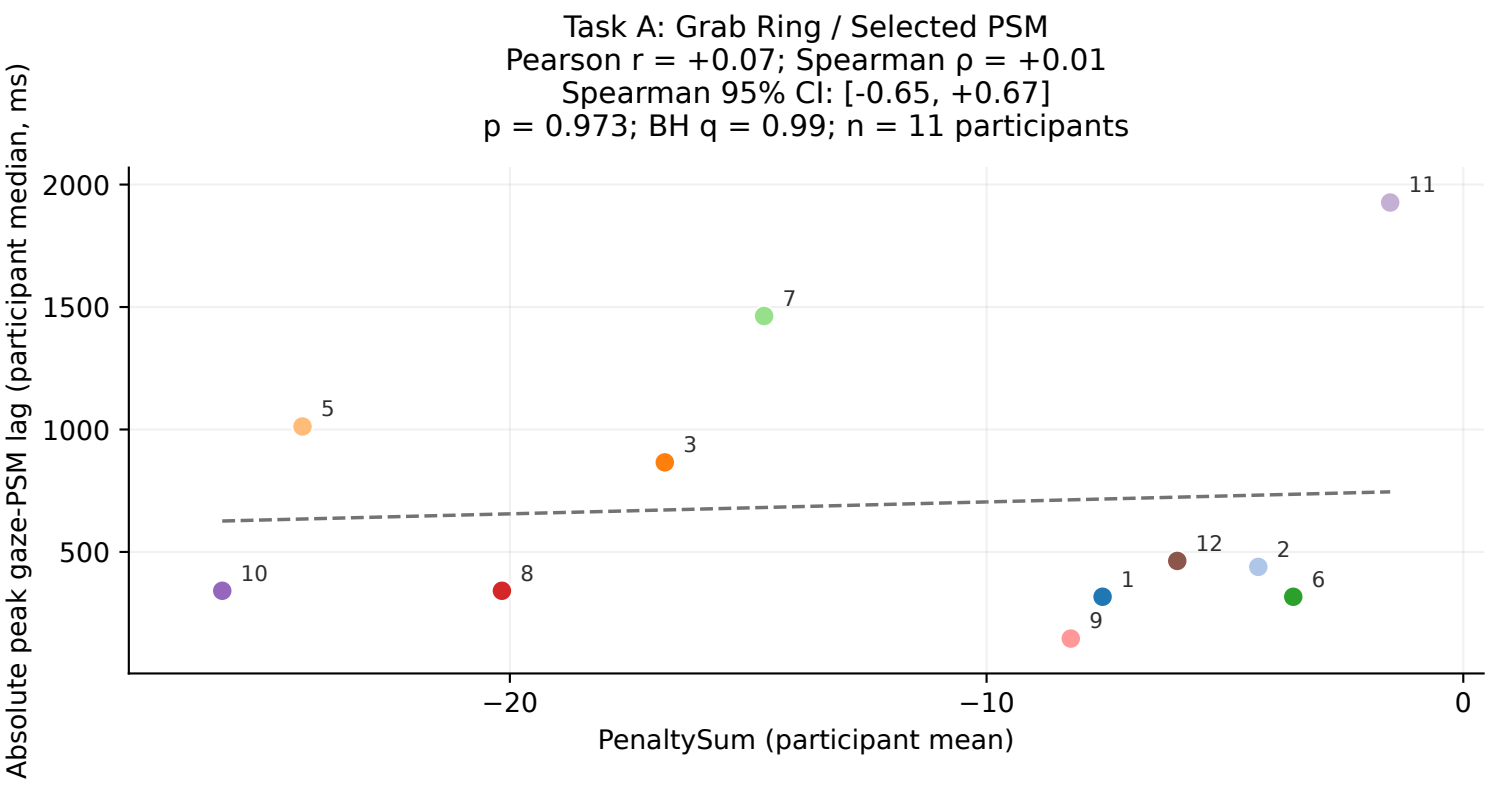
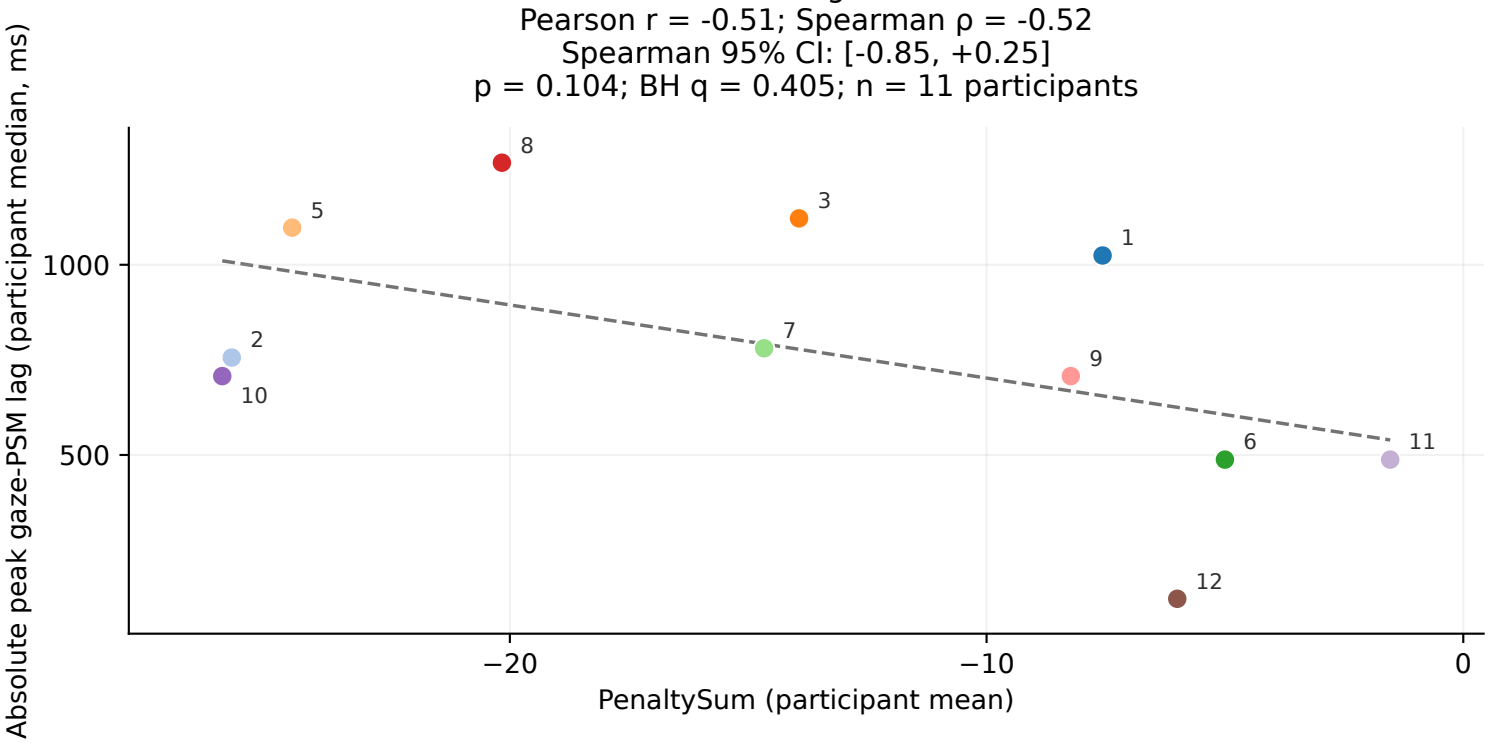
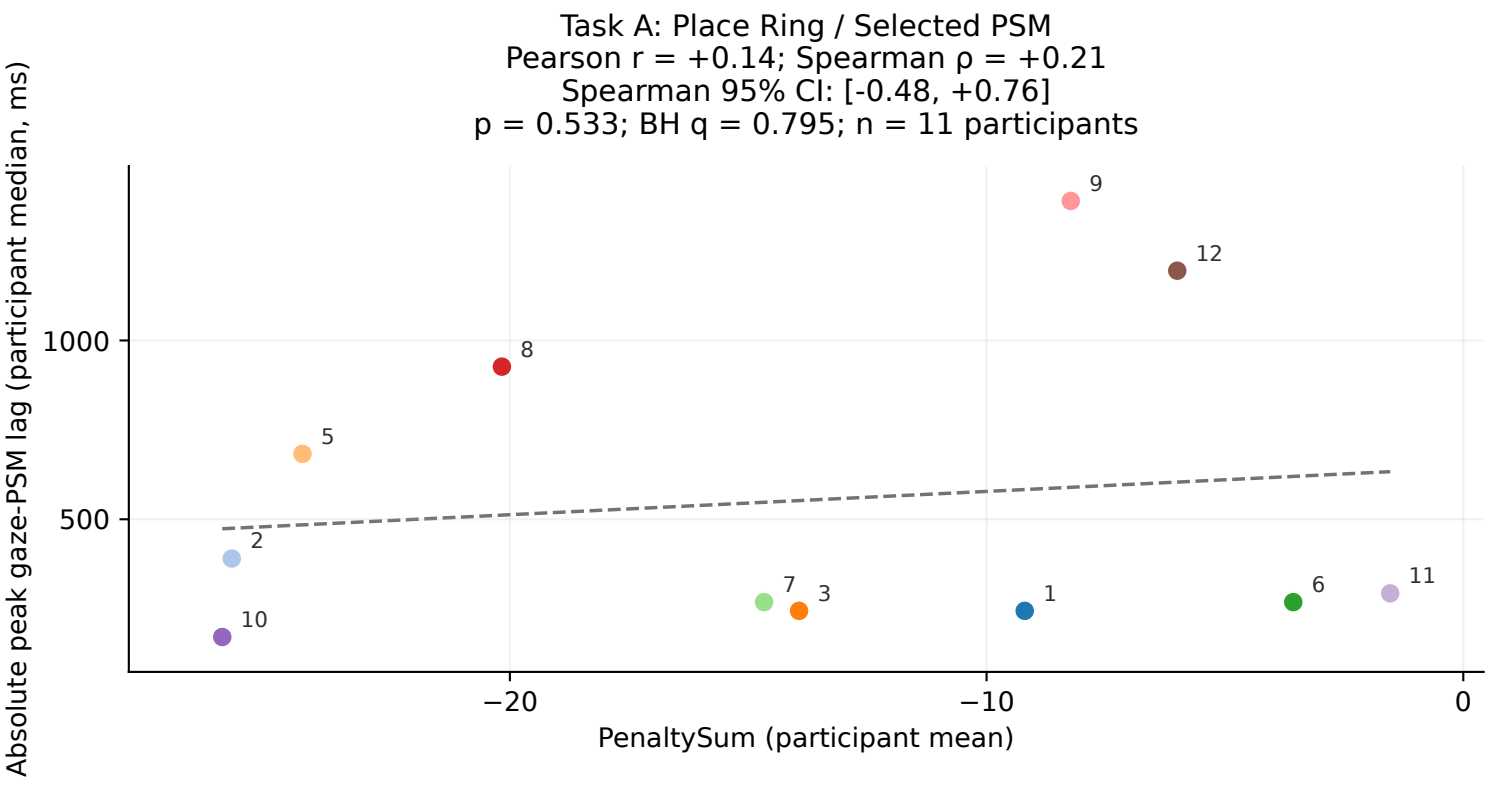
Task A: valid-phase absolute peak lag versus PenaltySum Position (primary analysis)



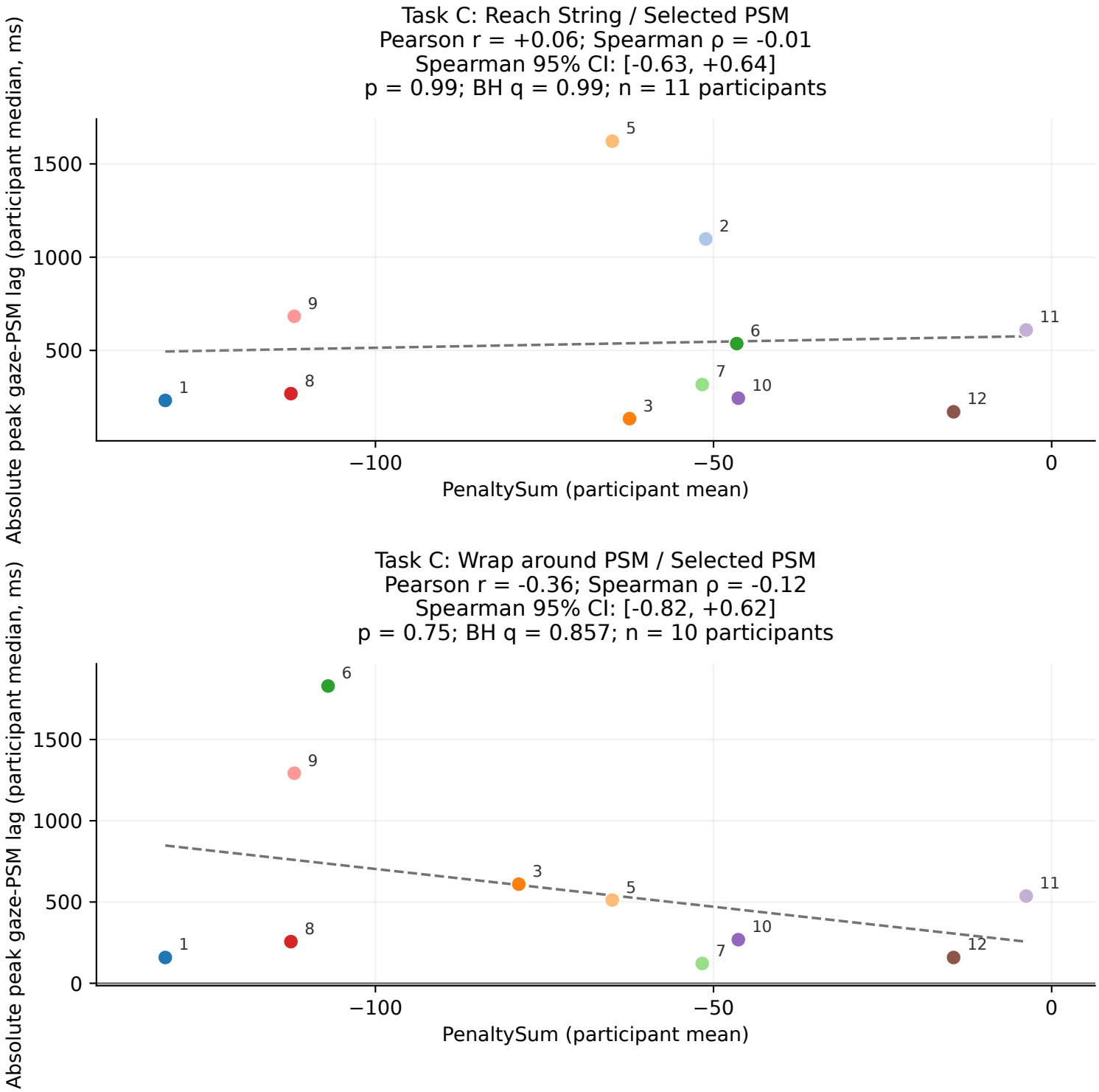






One point per participant; labels show participant IDs. Each point uses the same eligible attempts for PenaltySum and absolute peak lag.
Spearman 95% CIs: exploratory participant percentile bootstrap, unadjusted. p: permutation; q: BH; Pearson r/fit: descriptive.
CI unreliable: <97.5% defined bootstrap draws. BH family spans phases and features; phase scores are whole-attempt.
PenaltySum uses the supplied column values directly; attempts with missing PenaltySum values are excluded.
One PSM selected per recording/stage by signed xcorr peak, before score matching.
Selection identifies stronger correlation, not the operated instrument. Axes vary by stage.

Task C: valid-phase absolute peak lag versus PenaltySum Position (primary analysis)



One point per participant; labels show participant IDs. Each point uses the same eligible attempts for PenaltySum and absolute peak lag.
Spearman 95% CIs: exploratory participant percentile bootstrap, unadjusted. p: permutation; q: BH; Pearson r/fit: descriptive.
CI unreliable: <97.5% defined bootstrap draws. BH family spans phases and features; phase scores are whole-attempt.
PenaltySum uses the supplied column values directly; attempts with missing PenaltySum values are excluded.
One PSM selected per recording/stage by signed xcorr peak, before score matching.
Selection identifies stronger correlation, not the operated instrument. Axes vary by stage.